

Student Name: _____

Teacher: _____



HURLSTONE AGRICULTURAL HIGH SCHOOL

2021

Year 12
TASK ONE

Mathematics Extension 2

Examiners Mr G Rawson, Mrs P Biczó, Mr G Huxley

**Total
marks: 33**

Section 1 – 3 marks

- Attempt Questions 1-3
- Allow about 4 minutes for this section

Section 2 – 30 marks

- Attempt Questions 4-6
- Allow about 36 minutes for this section

**General
Instructions**

- Reading time – 3 minutes
- Working time – 40 minutes
- Write using black pen.
- Calculators approved by NESA may be used.
- A reference sheet is provided.
- A multiple choice answer sheet is provided at the back of the paper.
- In Questions 4-6 show relevant mathematical reasoning and/or calculations.
- Write your name at the top of each booklet. Answer the questions in the space provided.
- Extra booklets are available if you require more space.

Written by 2021 HAHS Maths Faculty, rewritten by Ariq Abdullah

Section 1

3 marks

Attempt Questions 1-3

1. Given $z = 1 - i$, which expression is equal to z^3 ?

- a. $\sqrt{2}e^{-\frac{3\pi i}{4}}$
- b. $2\sqrt{2}e^{-\frac{3\pi i}{4}}$
- c. $\sqrt{2}e^{\frac{3\pi i}{4}}$
- d. $2\sqrt{2}e^{\frac{3\pi i}{4}}$

2. The complex number z satisfies $\arg\left(\frac{z-2}{z+2i}\right) = -\frac{\pi}{2}$.

What is the maximum value of $|z|$?

- a. $\sqrt{2}$
- b. $2\sqrt{2}$
- c. $2 - \sqrt{2}$
- d. $2 + \sqrt{2}$

3. Which of the following is equivalent to $\left(\frac{e^{-\frac{7i\pi}{6}}}{e^{i\pi}}\right)$?

- a. $\frac{1}{2} + \frac{\sqrt{3}}{2}i$
- b. $\frac{1}{2} + \frac{\sqrt{3}}{2}i$
- c. $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$
- d. $\frac{\sqrt{3}}{2} - \frac{1}{2}i$

Section 2

30 marks

Attempt Questions 4-6

Allow about 36 minutes for this section

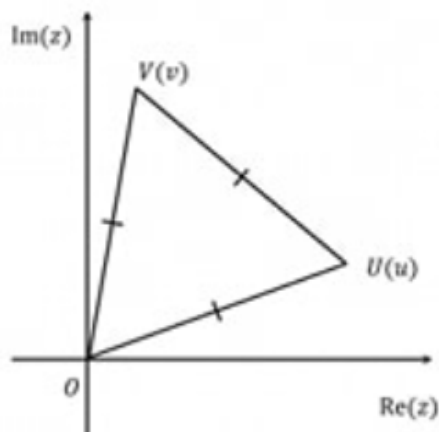
Question 4 (10 marks)	Marks
(a) Let $z = 4 + i$ and $w = \bar{z}$. Find, in the form $x + iy$,	
(i) w	1
(ii) $w - z$	1
(b) Let $\alpha = 1 + i\sqrt{3}$ and $\beta = 1 + i$.	
(i) Find $\frac{\alpha}{\beta}$ in the form $x + iy$	1
(ii) Express α in modulus-argument form.	2
(iii) Given that β has the modulus-argument form	1
$\beta = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$	
Find the modulus-argument form of $\frac{\alpha}{\beta}$.	
(iv) Hence, find the exact value of $\sin \frac{\pi}{12}$.	1
(c) Let $z = 2(\cos \theta + i \sin \theta)$.	
(i) Find $\overline{1 - z}$.	1
(ii) Show that the real part of $\frac{1}{1-z}$ is $\frac{1-2\cos\theta}{5-4\cos\theta}$.	2
Question 5 (10 marks)	Marks
(a)	
(i) Express $-\sqrt{3} - i$ in modulus-argument form.	2
(ii) De Moivre's theorem states that $(rcis\theta)^n = r^n cis n\theta$ for any integer n . Show that $(-\sqrt{3} - i)^6$ is a real number.	2
(b) Sketch the region in the complex plane where the inequalities $1 \leq z \leq 2$ and $0 \leq z + \bar{z} \leq 3$ hold simultaneously.	3
(c) Let $z = 2(\cos \theta + i \sin \theta)$.	3
Question 6 (10 marks)	Marks
(a) Given that $z = 3 + i$ is a root of $z^2 + pz + q = 0$, where p and q are real, find the values of p and q .	3
(b) Solve the equation $ e^{2i\theta} + e^{-2i\theta} = 1$ where $-\pi \leq \theta \leq \pi$.	3

Question 6 continued on next page

- (c) Let the complex numbers $V(v)$, $U(u)$ and the origin form a triangle of area $\frac{1}{2}u^2$ on the Argand Diagram.

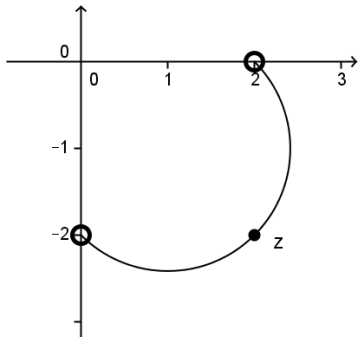
M is the midpoint of UV.

Let $\angle UMO = \theta$



- (i) Show that $|u + v||u - v| \sin \theta = 4$. **2**
- (ii) If $\theta = \frac{\pi}{2}$, prove that $(|u| - |v|)(|u| + |v|) = 0$. **2**

End of Exam

Year 12	Mathematics Extension 2	Ass Task 1 2021 HSC
MULTIPLE CHOICE	Solutions and Marking Guidelines	
Outcome	Solutions	Marking Guidelines
MEX12-4	1. $1-i = \sqrt{1^2 + (-1)^2} = \sqrt{2}$ $\arg(1-i) = -\frac{\pi}{4}$ $\therefore 1-i = \sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right)$ $(1-i)^3 = (\sqrt{2})^3 \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right)^3$ $= 2\sqrt{2} \left(\cos\left(-\frac{3\pi}{4}\right) + i \sin\left(-\frac{3\pi}{4}\right) \right)$ $= 2\sqrt{2} e^{\frac{3\pi i}{4}} \quad \therefore B$	1 mark
MEX12-4	2. $\arg\left(\frac{z-2}{z+2i}\right) = -\frac{\pi}{2}$ $\therefore \arg(z-2) - \arg(z-(-2i)) = -\frac{\pi}{2}$, and so z lies on a semicircle (excluding the endpoints) whose diameter is the line joining the points 2 and $-2i$ on the argand diagram. Need to test whether top or bottom half of the semicircle. Testing if z is at $(0, 0)$: $\arg(z-2) - \arg(z+2i) = 180^\circ - 90^\circ \neq -\frac{\pi}{2}$. If z is $(2, -2)$, $\arg(z-2) - \arg(z+2i) = -90^\circ - 0^\circ = -\frac{\pi}{2}$. $\therefore z$ is on the semicircle containing the point $(2, -2)$. From the diagram, the position of z that gives the maximum value of z is when z is at $(2, -2)$.  $\therefore z$ is $d(0,0), (2,-2) = \sqrt{8}$ $\therefore B$	1 mark

MEX12-4	$ \begin{aligned} 3. e^{-\left(\frac{7i\pi}{6}\right)} \div e^{i\pi} &= e^{-\left(\frac{7i\pi}{6}\right) - i\pi} \\ &= e^{-i\left(\frac{7\pi+6\pi}{6}\right)} \\ &= e^{-\frac{13i\pi}{6}} \\ &= \cos\left(-\frac{13\pi}{6}\right) + i\sin\left(-\frac{13\pi}{6}\right) \\ &= \cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right) \\ &= \cos\frac{\pi}{6} - i\sin\frac{\pi}{6} \\ &= \frac{\sqrt{3}}{2} - \frac{1}{2}i \end{aligned} $ $\therefore D$	1 mark
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Year 12	Mathematics Extension 2	Ass Task 1 2021 HSC
Question No. 4	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
MEX12-4 uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems		
Part / Outcome	Solutions	Marking Guidelines
(a)(i)	$w = \bar{z}$ $= \overline{4 + i}$ $= 4 - i$	<u>1 mark</u> : correct solution
(a)(ii)	$w - z = (4 - i) - (4 + i)$ $= -2i$	<u>1 mark</u> : correct solution
(b)(i)	$\frac{\alpha}{\beta} = \frac{1 + i\sqrt{3}}{1 + i} \times \frac{1 - i}{1 - i}$ $= \frac{1 - i + i\sqrt{3} + \sqrt{3}}{1 + 1}$ $= \frac{(1 + \sqrt{3}) + (\sqrt{3} - 1)i}{2}$ $= \left(\frac{1 + \sqrt{3}}{2} \right) + \left(\frac{\sqrt{3} - 1}{2} \right) i$	<u>1 mark</u> : correct solution <i>NB: the question asked for this to be in the form $x + iy$, which really means the real and imaginary parts should be <u>fully separated</u></i>
(b)(ii)	$\alpha = 1 + i\sqrt{3}$ $ \alpha = \sqrt{1 + 3} = 2$ $\arg(\alpha) = \tan^{-1} \left(\frac{\sqrt{3}}{1} \right) = \frac{\pi}{3}$ $\therefore \alpha = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$	<u>2 marks</u> : correct solution <u>1 mark</u> : substantially correct solution <i>NB: if using degrees, the degree symbol ($^{\circ}$) MUST be used. If no symbol is used, you are automatically referring to radians. 60 radians is not the correct angle here!</i>
(b)(iii)	$\frac{\alpha}{\beta} = \frac{2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)}{\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)}$ $= \frac{2}{\sqrt{2}} \left(\cos \left(\frac{\pi}{3} - \frac{\pi}{4} \right) + i \sin \left(\frac{\pi}{3} - \frac{\pi}{4} \right) \right)$ $= \sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$	<u>1 mark</u> : correct solution

Question 4 continued...

(b)(iv)

Comparing (i) and (iii), and equating imaginary parts:

$$\sqrt{2} \sin \frac{\pi}{12} = \frac{\sqrt{3}-1}{2}$$

$$\sin \frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

1 mark : correct solution

(c)(i)
MEX12-4

$$1-z = 1-2\cos\theta - 2i\sin\theta$$

$$\overline{1-z} = (1-2\cos\theta) + 2i\sin\theta$$

1 mark : correct solution

(c)(ii)

$$\frac{1}{1-z} = \frac{1}{1-z} \times \frac{\overline{1-z}}{\overline{1-z}}$$

$$= \frac{1-z}{|1-z|^2}$$

$$= \frac{(1-2\cos\theta) + 2i\sin\theta}{(1-2\cos\theta)^2 + 4\sin^2\theta}$$

$$= \frac{(1-2\cos\theta) + 2i\sin\theta}{1-4\cos\theta + 4\cos^2\theta + 4\sin^2\theta}$$

$$= \frac{(1-2\cos\theta) + 2i\sin\theta}{1-4\cos\theta + 4}$$

$$= \frac{(1-2\cos\theta) + 2i\sin\theta}{5-4\cos\theta}$$

$$\operatorname{Re}\left(\frac{1}{1-z}\right) = \frac{1-2\cos\theta}{5-4\cos\theta}$$

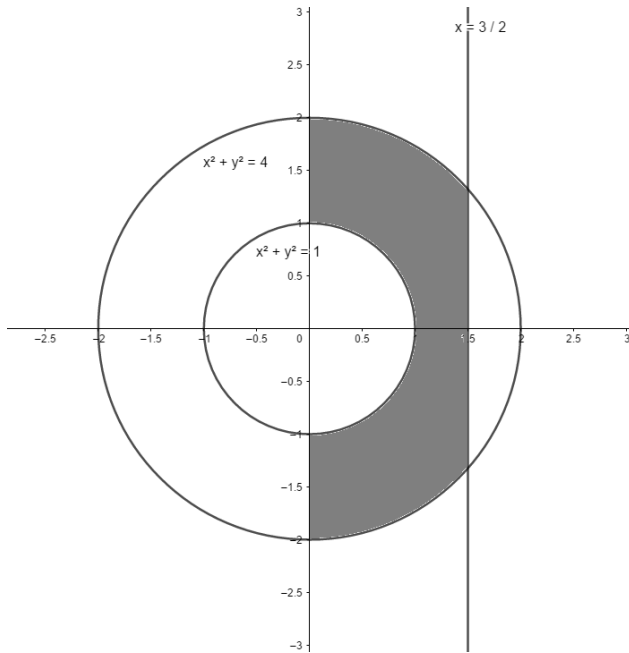
2 marks : correct solution

1 mark : substantially correct solution

NB: as this is a 'show' question, it should have been clearly stated that $\sin^2x + \cos^2x = 1$ was being used to help simplify the denominator

Outcomes Addressed in this Question

MEX12-4 uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems

Outcome	Solutions	Marking Guidelines
(a) MEX12-4	(i) $-\sqrt{3} - i = 2\text{cis}\left(-\frac{5\pi}{6}\right)$ (ii) $\begin{aligned} (-\sqrt{3} - i)^6 &= 2^6 \text{cis}\left(-\frac{5\pi}{6} \times 6\right) \\ &= 2^6 \text{cis}(-5\pi) \\ &= 2^6(-1) = -2^6, \text{ which is real.} \end{aligned}$	(a) (i) 2 marks: Both modulus and principal argument correct. 1 mark: 1 correct component. (ii) 2 marks: Correct use of DeMoivre's Theorem and correct real number 1 mark: 1 component correct.
(b) MEX12-4	$0 \leq z + \bar{z} \leq 3 \rightarrow 0 \leq 2\text{Re}(z) \leq 3 \rightarrow 0 \leq x \leq \frac{3}{2}$ $1 \leq z \leq 2$ represents the area between concentric circles. 	(b) (i) 3 marks: Complete solution. 2 marks: One error in solving method. 1 mark: Some relevant progress.
(c) MEX12-4	$\begin{aligned} z &= \sqrt{(1 - \cos 2\theta)^2 + (\sin 2\theta)^2} \\ &= \sqrt{1 + 2\cos 2\theta + \cos^2 2\theta + \sin^2 2\theta} \\ &= \sqrt{2 - 2\cos 2\theta} \\ &= \sqrt{2(1 - (1 - 2\sin^2 \theta))} \\ &= \sqrt{4\sin^2 \theta} \\ &= 2\sin \theta \end{aligned}$	(c) (i) 3 marks: Correct proof, showing all relevant steps 2 marks: One error in solving method. 1 mark: Some relevant progress.

Year 12	Mathematics Extension 2	Task 1 2021
Question No. 6	Solutions and Marking Guidelines	
Outcomes Addressed in this Question		
MEX12-4 uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems		
Outcome	Solutions	Marking Guidelines
MEX12-4	(a) As real coefficients, complex roots must occur in conjugate pairs. $\therefore$ as $3+i$ is a root, $3-i$ is a root. Sum of roots of $z^2 + pz + q = 0$ is $\frac{-p}{1}$ and product of the roots is $\frac{q}{1}$. $\therefore 3+i+3-i = -p$ and $(3+i)(3-i) = q$ $\therefore p = -6$ and $q = 10$.	3 marks: correct solution 2 marks: partially correct solution 1 mark: finds either p or q or the other root
MEX12-4	(b) $e^{2i\theta} + e^{-2i\theta} = 1$ $\therefore \cos 2\theta + i \sin 2\theta + \cos(-2\theta) + i \sin(-2\theta) = 1$ $\therefore \cos 2\theta + i \sin 2\theta + \cos 2\theta - i \sin 2\theta = 1$ $\therefore 2 \cos 2\theta = 1$ $\therefore \cos 2\theta = \frac{1}{2}$ $\therefore \cos 2\theta = \pm \frac{1}{2}$ All quadrants; basic angle 60°; required to solve for $-\pi \leq \theta \leq \pi$, hence $-2\pi \leq 2\theta \leq 2\pi$. $\therefore 2\theta = \frac{\pi}{3}, \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, \frac{\pi}{3} - 2\pi, -\pi - \frac{\pi}{3}, -\pi + \frac{\pi}{3}, -\frac{\pi}{3}$ $\therefore 2\theta = \pm \frac{\pi}{3}, \pm \frac{2\pi}{3}, \pm \frac{4\pi}{3}, \pm \frac{5\pi}{3}$ $\therefore \theta = \pm \frac{\pi}{6}, \pm \frac{\pi}{3}, \pm \frac{2\pi}{3}, \pm \frac{5\pi}{6}$	3 marks: correct solution 2 marks: partially correct solution 1 mark: uses Euler's formula and simplifies

MEX12-4	<i>Question 6 continued...</i> (c) (i) Completing the parallelogram $OWPZ$, $\overrightarrow{OW} = w$, $\overrightarrow{OZ} = z$, $\overrightarrow{OP} = w + z$, $\overrightarrow{WZ} = z - w$. As the diagonals of a parallelogram bisect one another, $\overrightarrow{OM} = \frac{1}{2}(z + w)$ and $\overrightarrow{MZ} = \frac{1}{2}(z - w)$ Also, as triangles OMW and OMZ have the same base length and same perpendicular height, their areas are equal. $\therefore \text{area } \triangle OMZ = \frac{1}{2}$, given $\text{area } \triangle OWZ = 1$. $\therefore \frac{1}{2} \times OM \times MZ \times \sin \theta = \frac{1}{2}$ (using $\frac{1}{2}ab \sin C$ in $\triangle OMZ$) $\therefore \frac{1}{2} \times \left \frac{1}{2}(z + w) \right \times \left \frac{1}{2}(z - w) \right \times \sin \theta = \frac{1}{2}$ $\therefore \frac{1}{8} \times z + w \times z - w \times \sin \theta = \frac{1}{2}$ $\therefore z + w z - w \times \sin \theta = 4$. (ii) If $\theta = \frac{\pi}{2}$, then the parallelogram is a rhombus, hence all sides are equal. $\therefore z = w$ $\therefore z - w = 0$ $\therefore (z - w)(z + w) = 0 \times (z + w)$ $= 0$.	2 marks: correct solution 1 mark: partially correct solution 2 marks: correct solution 1 mark: partially correct solution
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